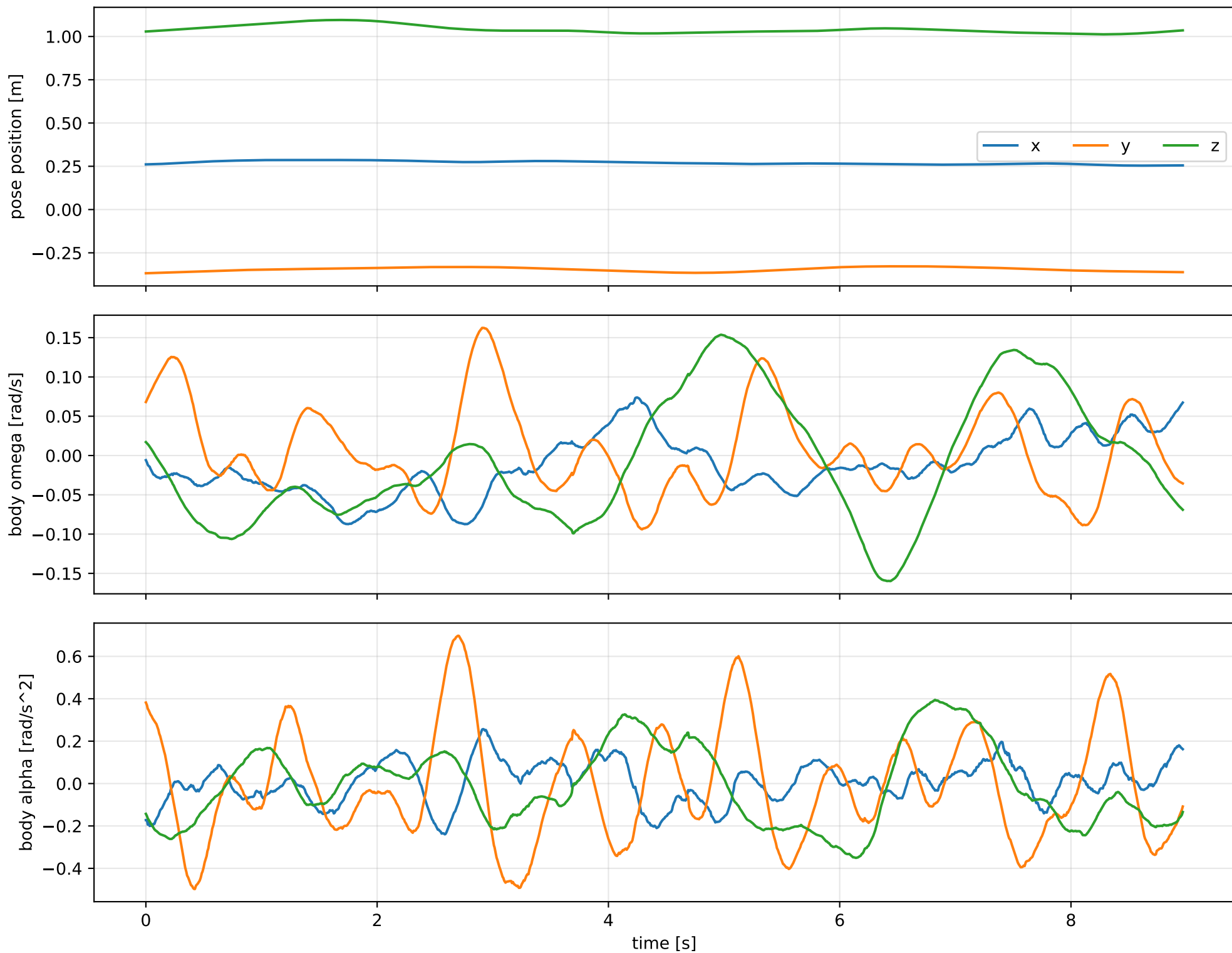
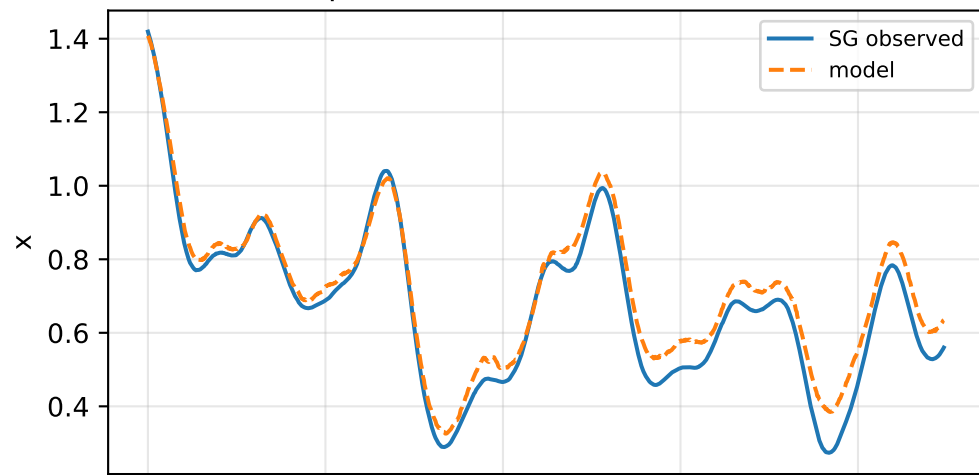


solver_standard_gauge_least_squares: SG trajectory and derived motion

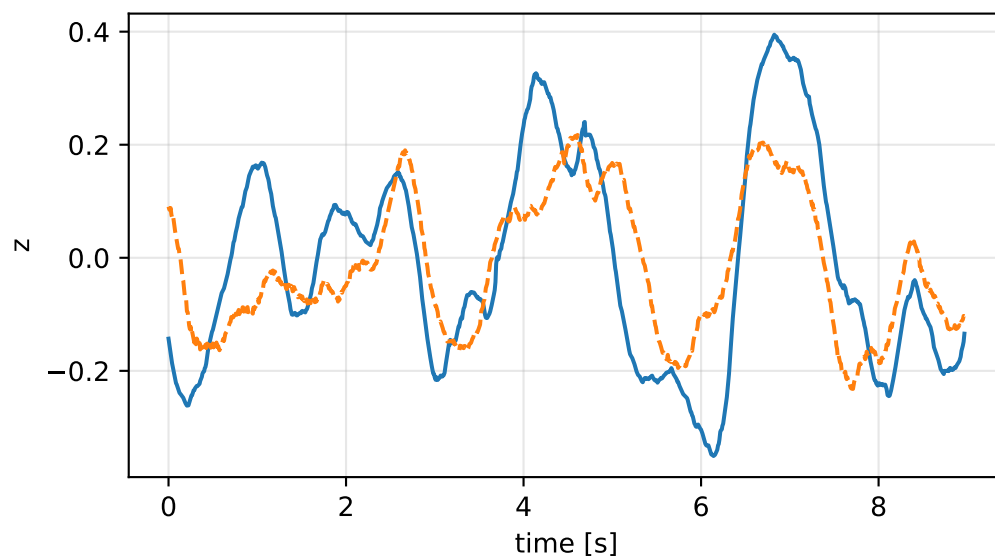
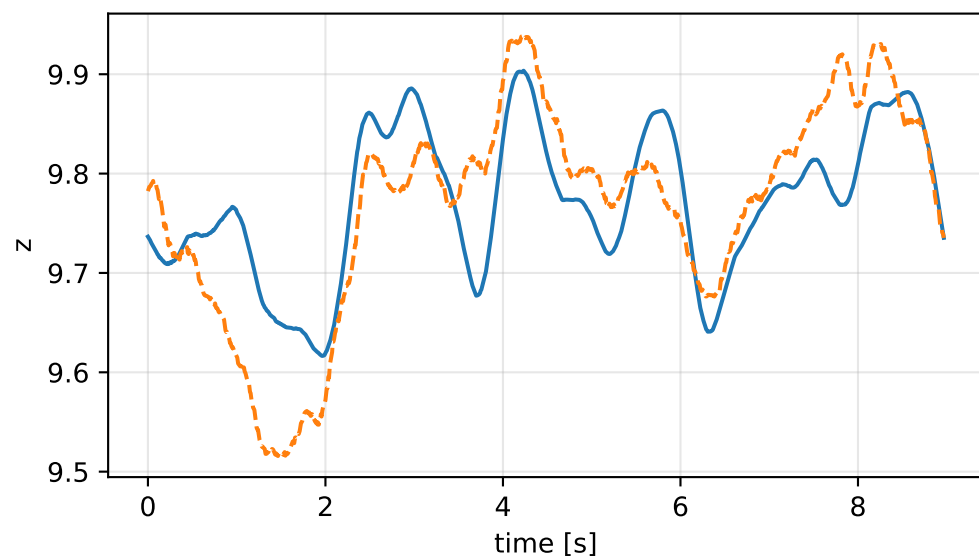
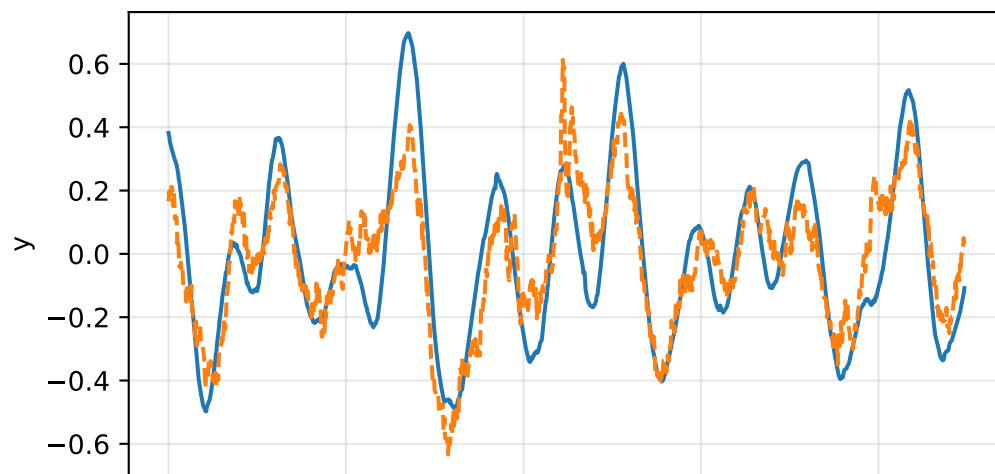
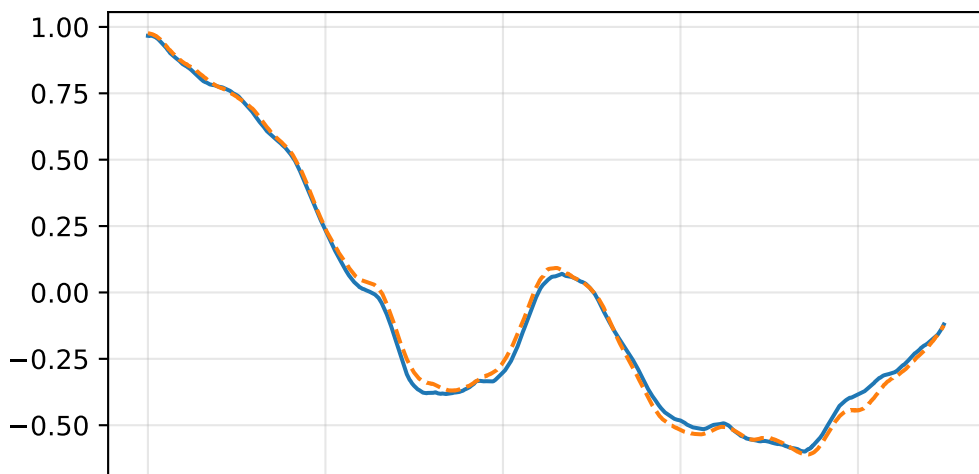
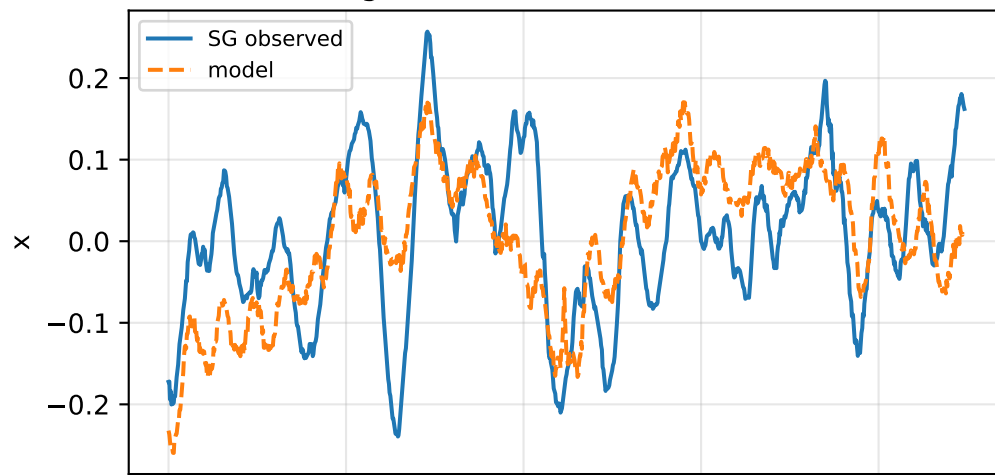


solver_standard_gauge_least_squares: acceleration residual objective

specific acceleration [m/s²]

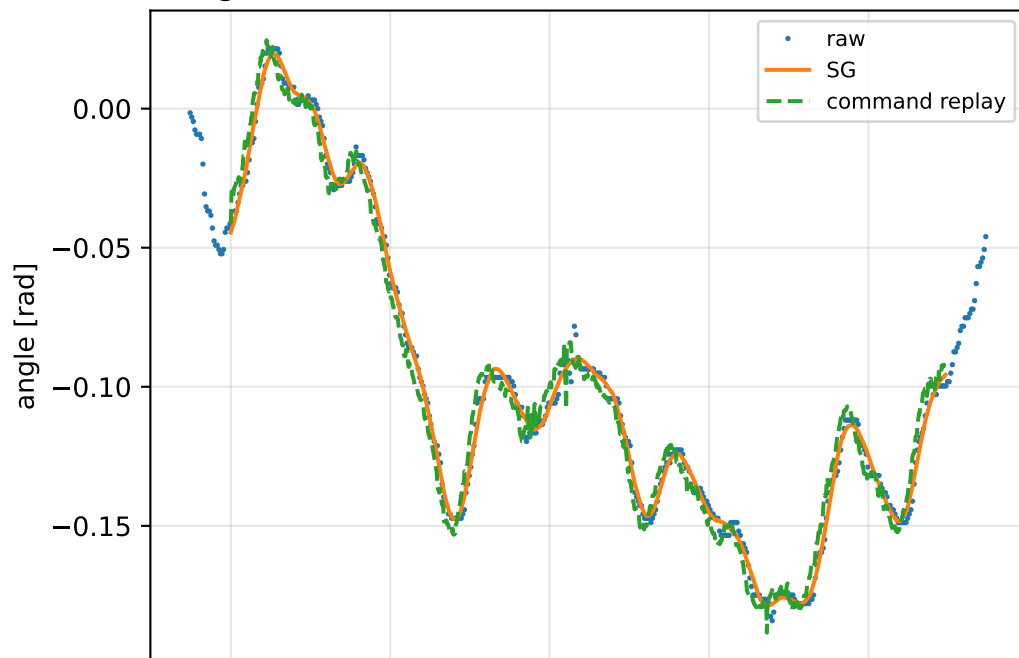


angular acceleration [rad/s²]

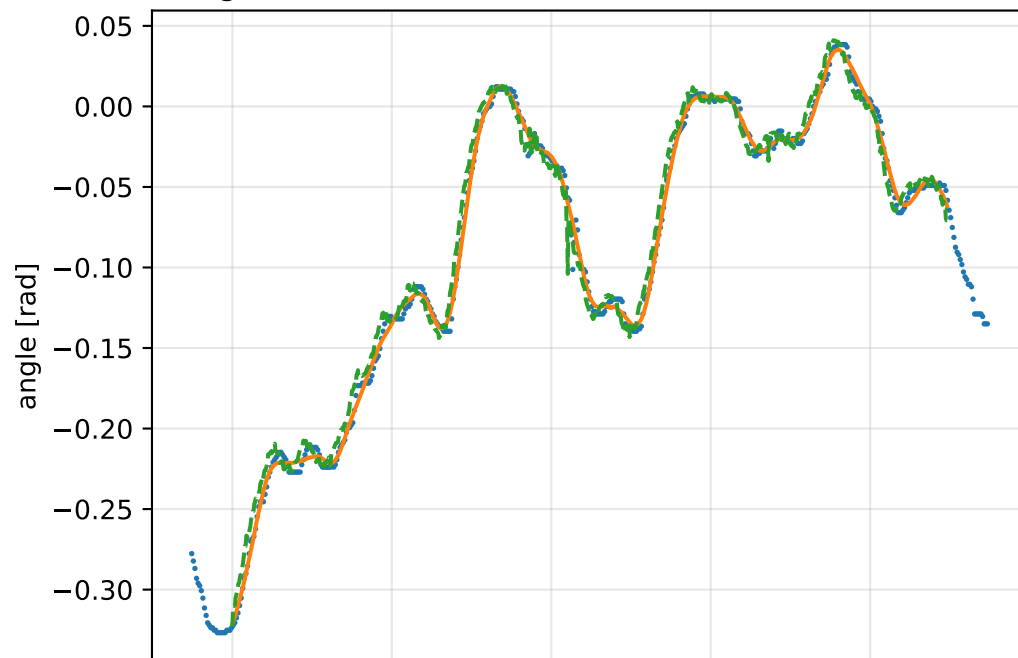


solver_standard_gauge_least_squares: gimbal measurement audit (objective=measured_sg)

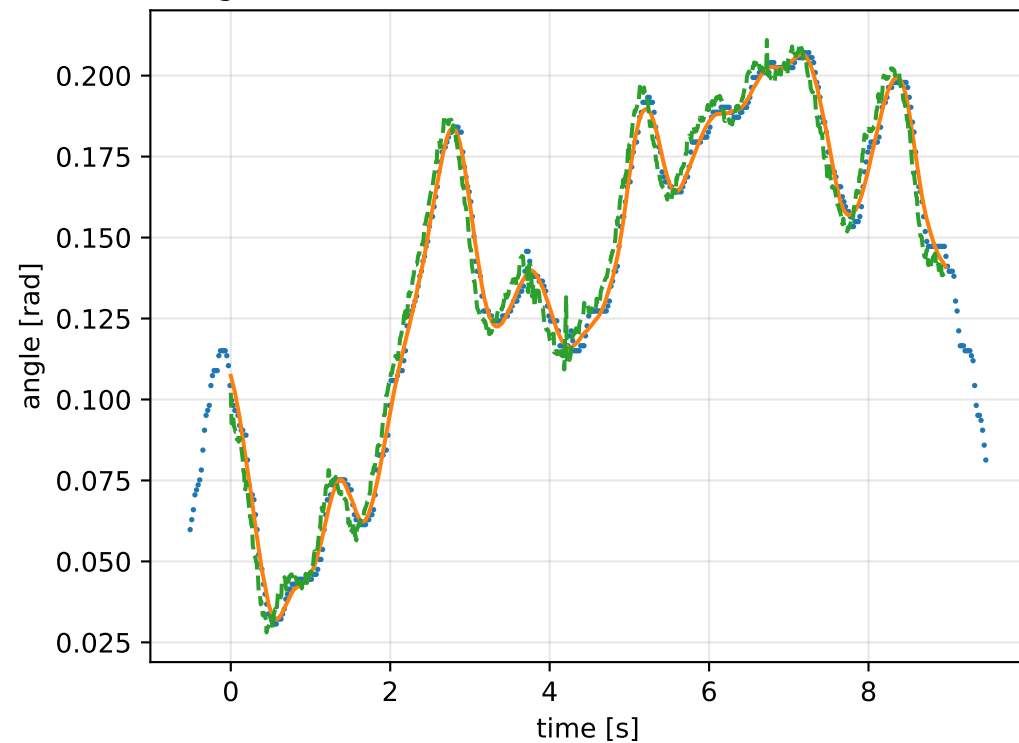
gimbal 1: RMSE=0.006345, max=0.01494 rad



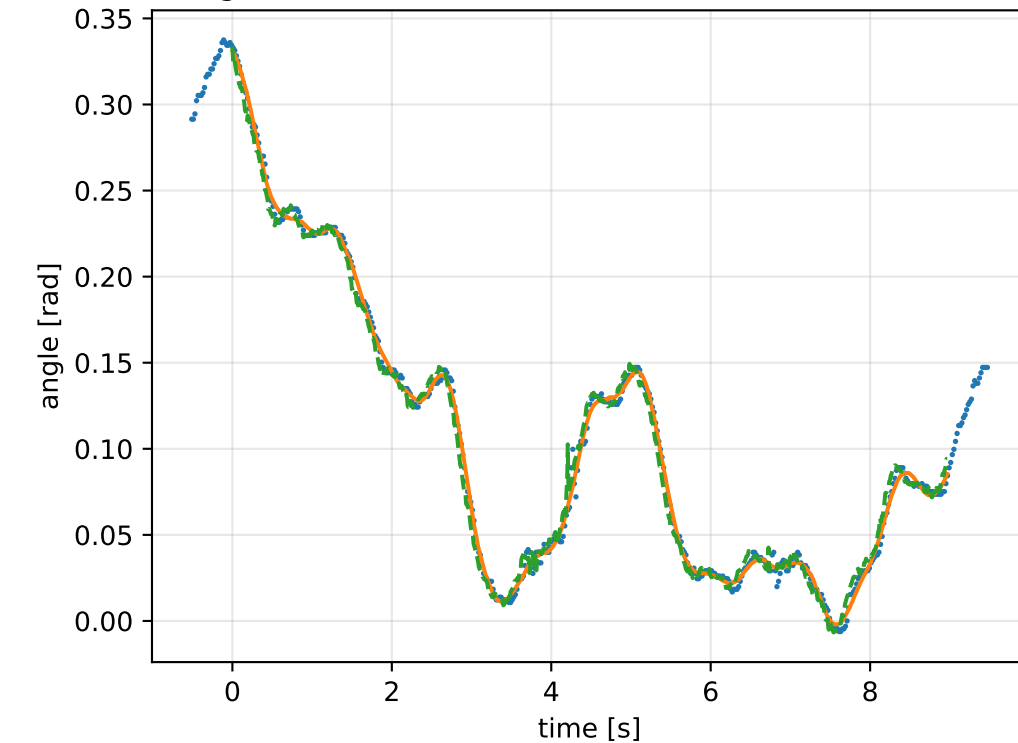
gimbal 2: RMSE=0.008835, max=0.04462 rad



gimbal 3: RMSE=0.007019, max=0.01611 rad

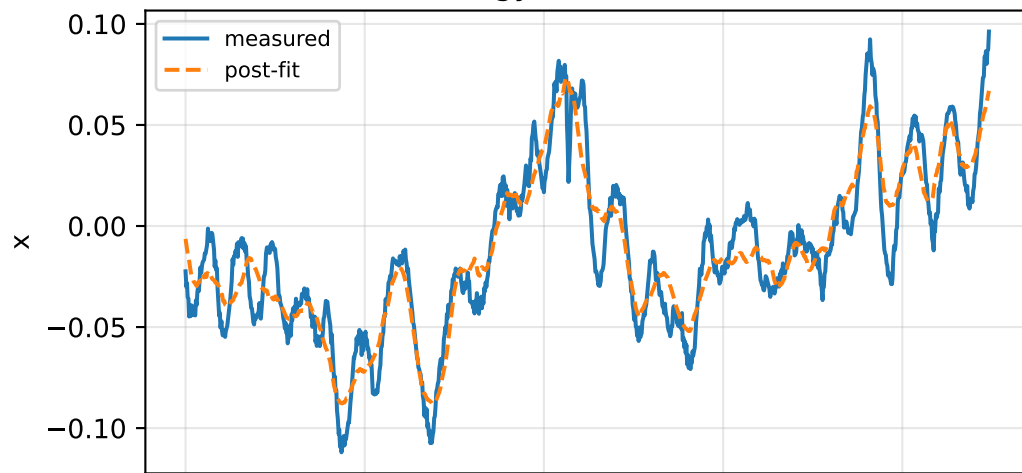


gimbal 4: RMSE=0.006321, max=0.03642 rad

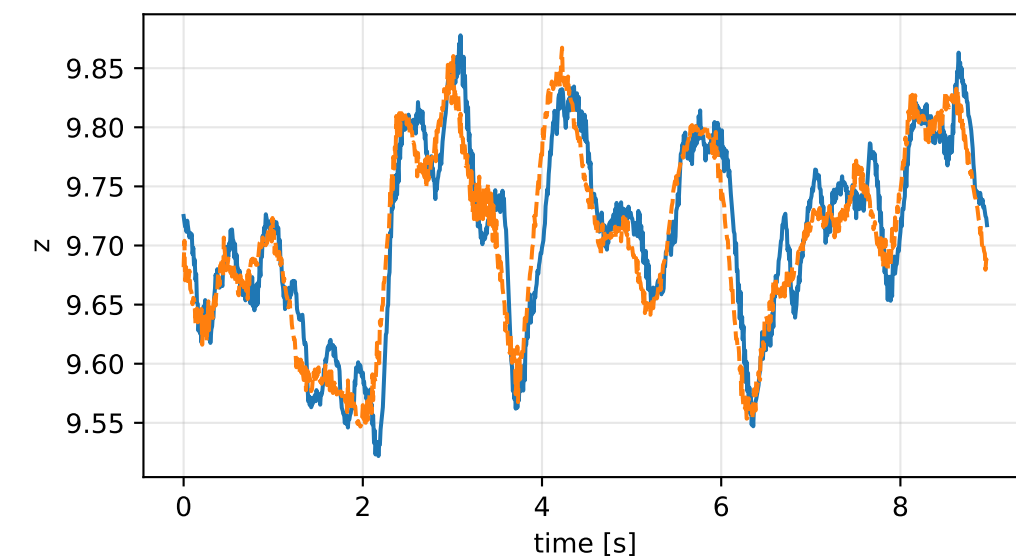
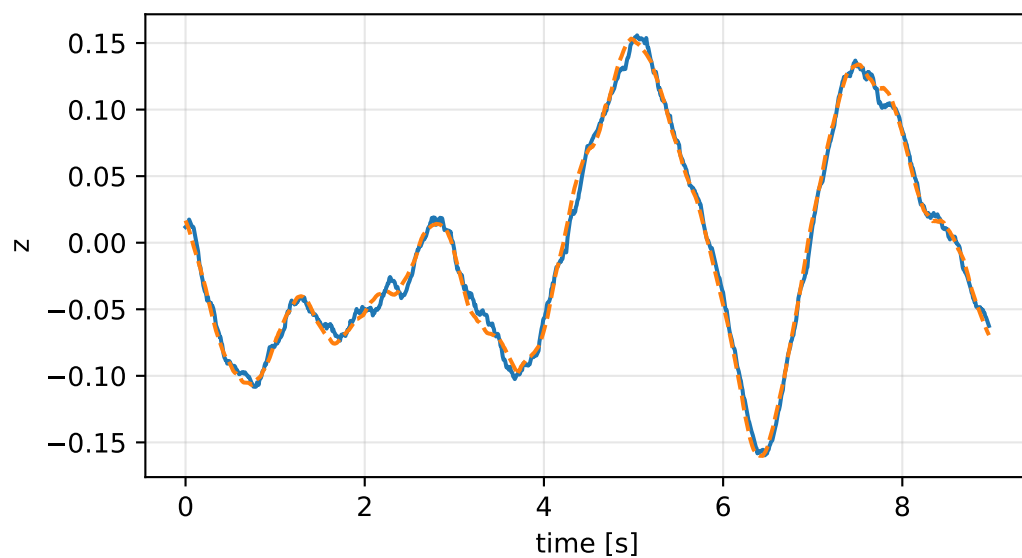
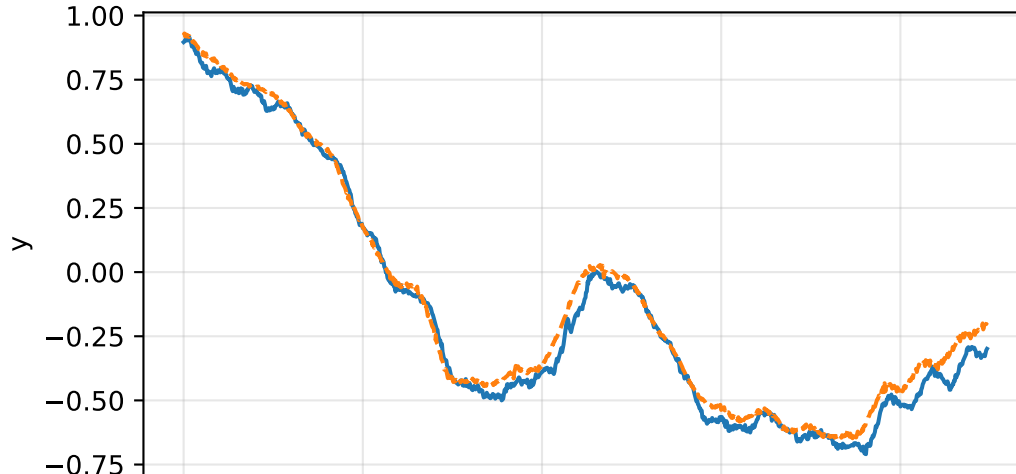
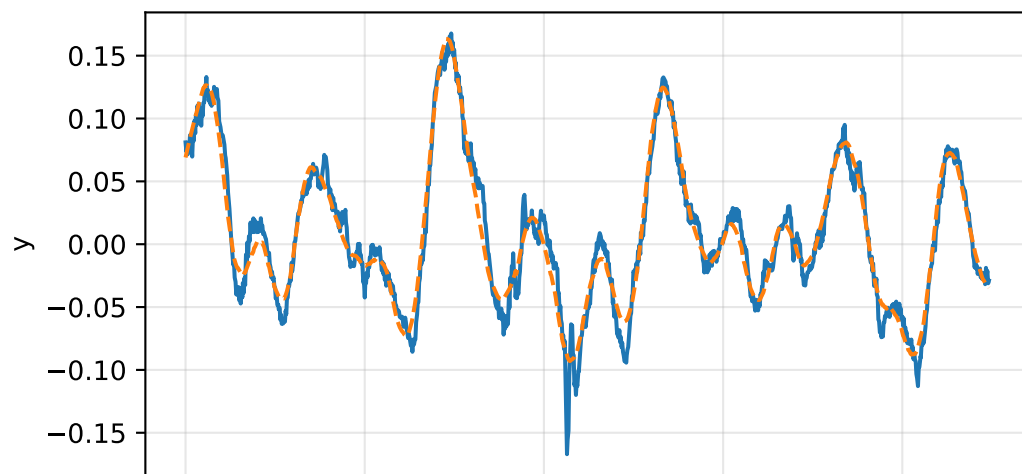
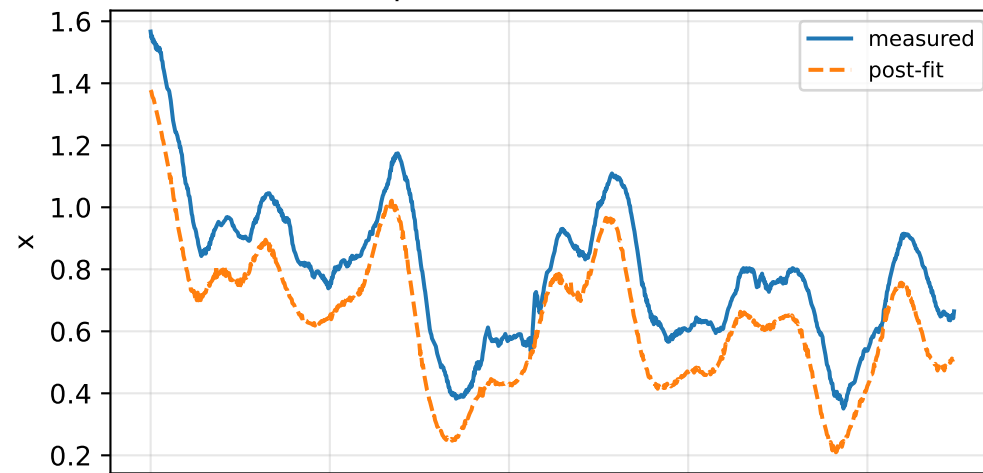


solver_standard_gauge_least_squares: IMU comparison (diagnostic only)

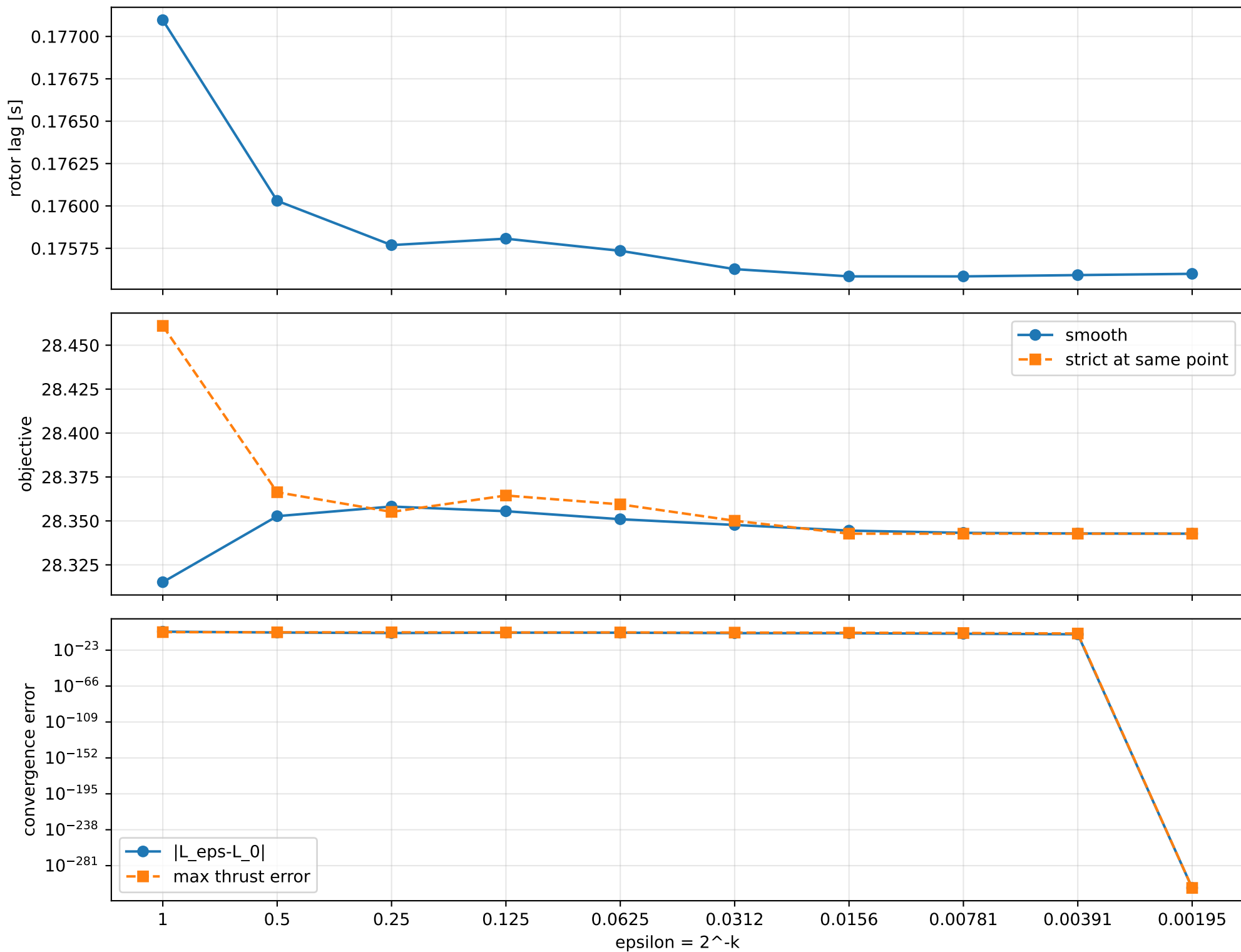
gyro [rad/s]



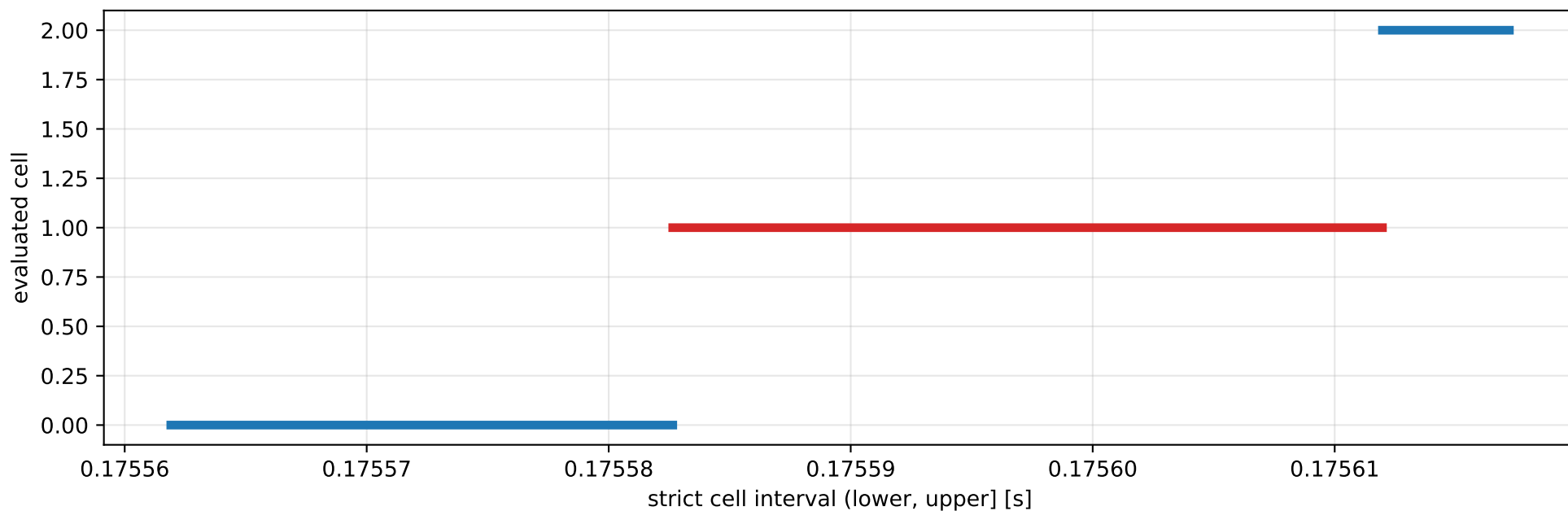
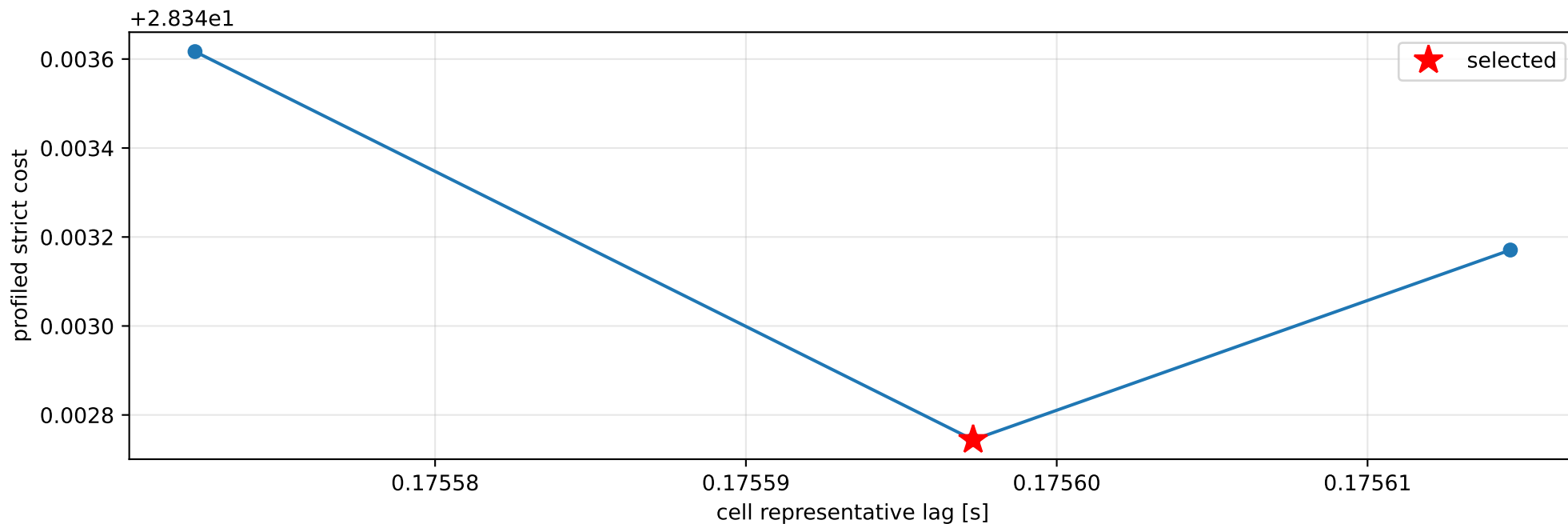
specific force [m/s^2]



solver_standard_gauge_least_squares: smooth-to-strict continuation



solver_standard_gauge_least_squares: exact strict-ZOH lag cells



selected (0.175582647, 0.175611973] s; final smooth 0.175599525 s; data boundary=False

solver_standard_gauge_least_squares: parameter estimate

Case: solver_standard_gauge_least_squares
status: completed (all stages completed)

mass [kg]: nominal 2.35159759; estimated 2.60279118e+66

inertia [kg m²]:

```
[[ 6.03652135e+65 -1.33873482e+64  4.69915778e+64]
 [-1.33873482e+64  2.85446945e+65 -7.95911248e+63]
 [ 4.69915778e+64 -7.95911248e+63  3.35490338e+65]]
```

CoG body [m]: [-0.01982309 -0.0551304 -0.00503022]

force effectiveness: [1.15702172e+66 1.13222336e+66 5.97356758e+65 7.25347975e+65]

scale-free J/m [m²]:

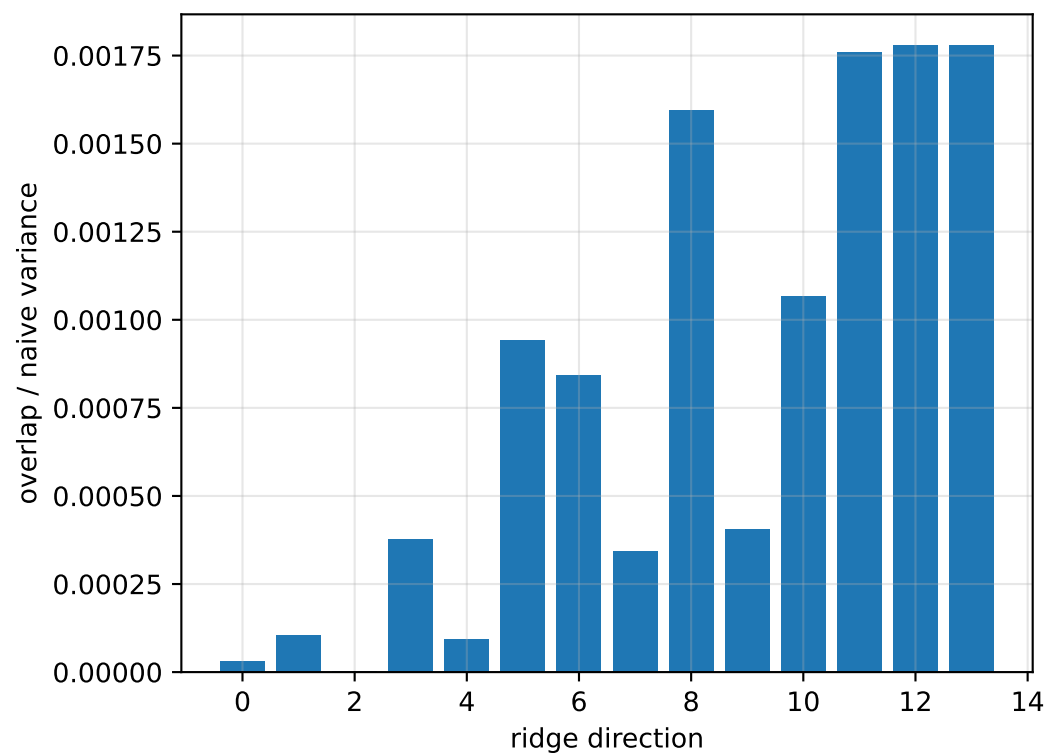
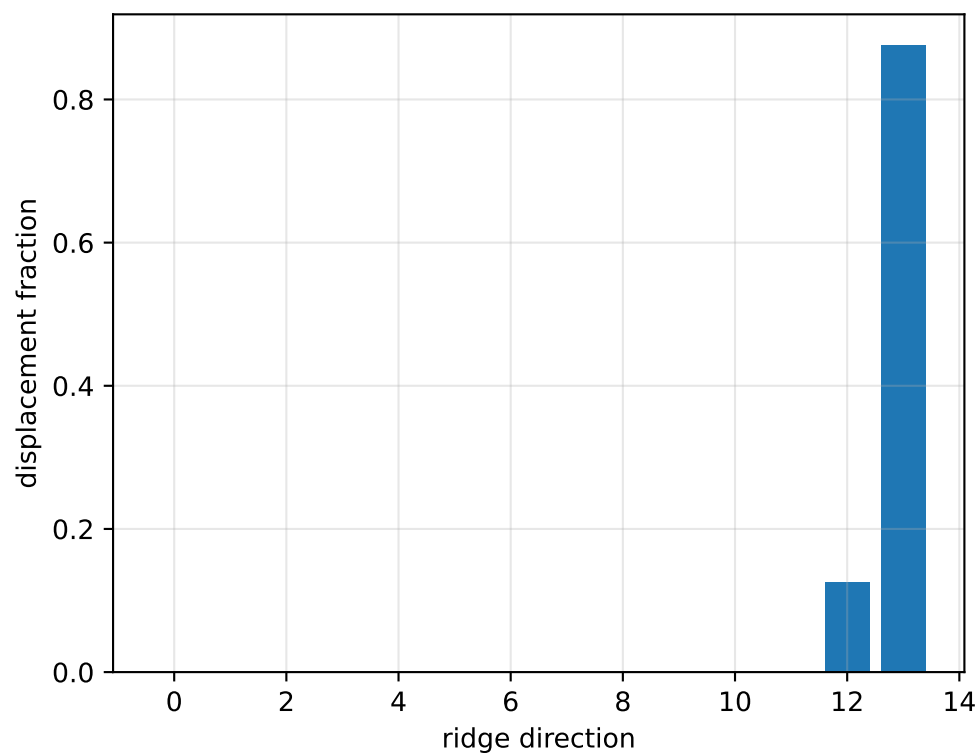
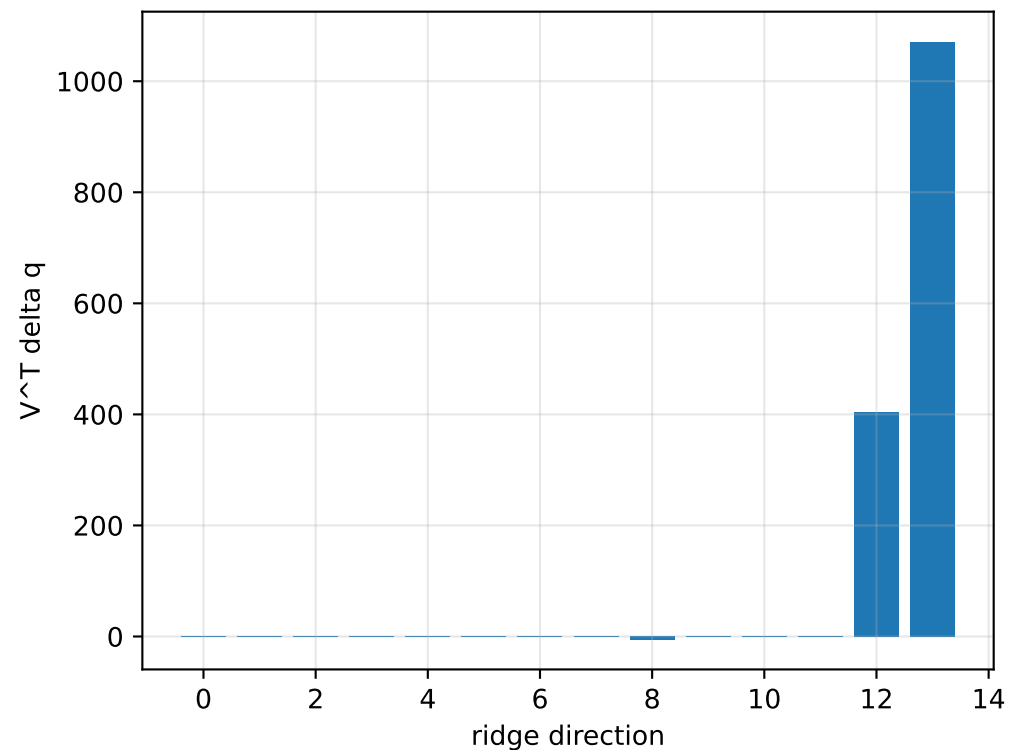
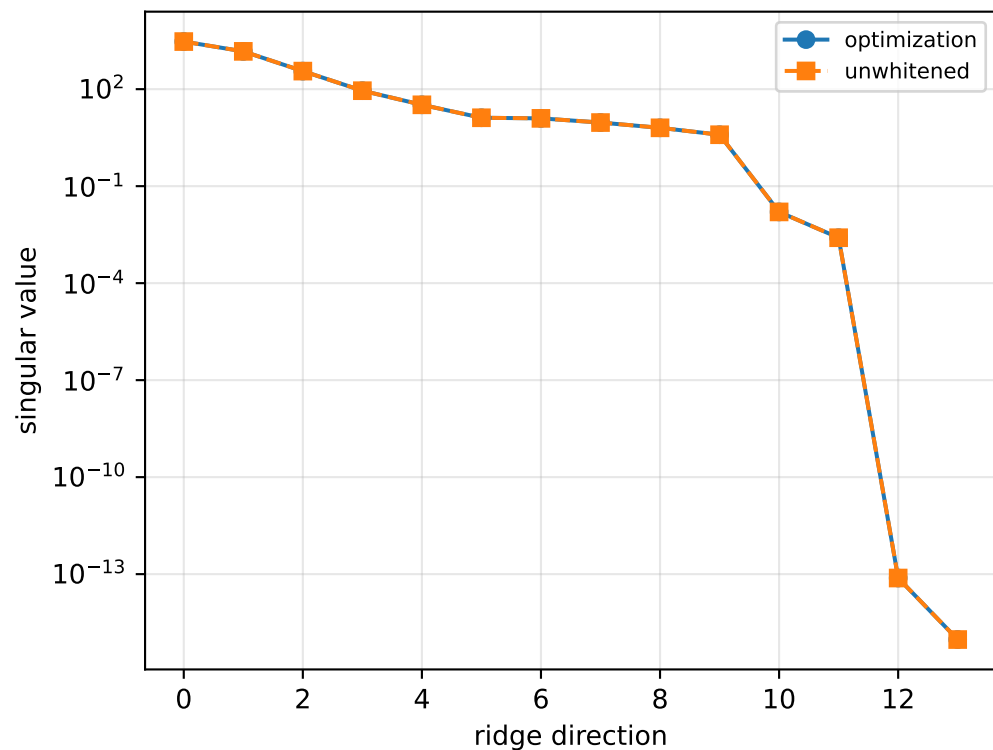
```
[[ 0.23192492 -0.00514346  0.0180543 ]
 [-0.00514346  0.10966955 -0.00305791]
 [ 0.0180543  -0.00305791  0.12889637]]
```

scale-free f/m: [0.44453114 0.43500353 0.22950622 0.27868082]

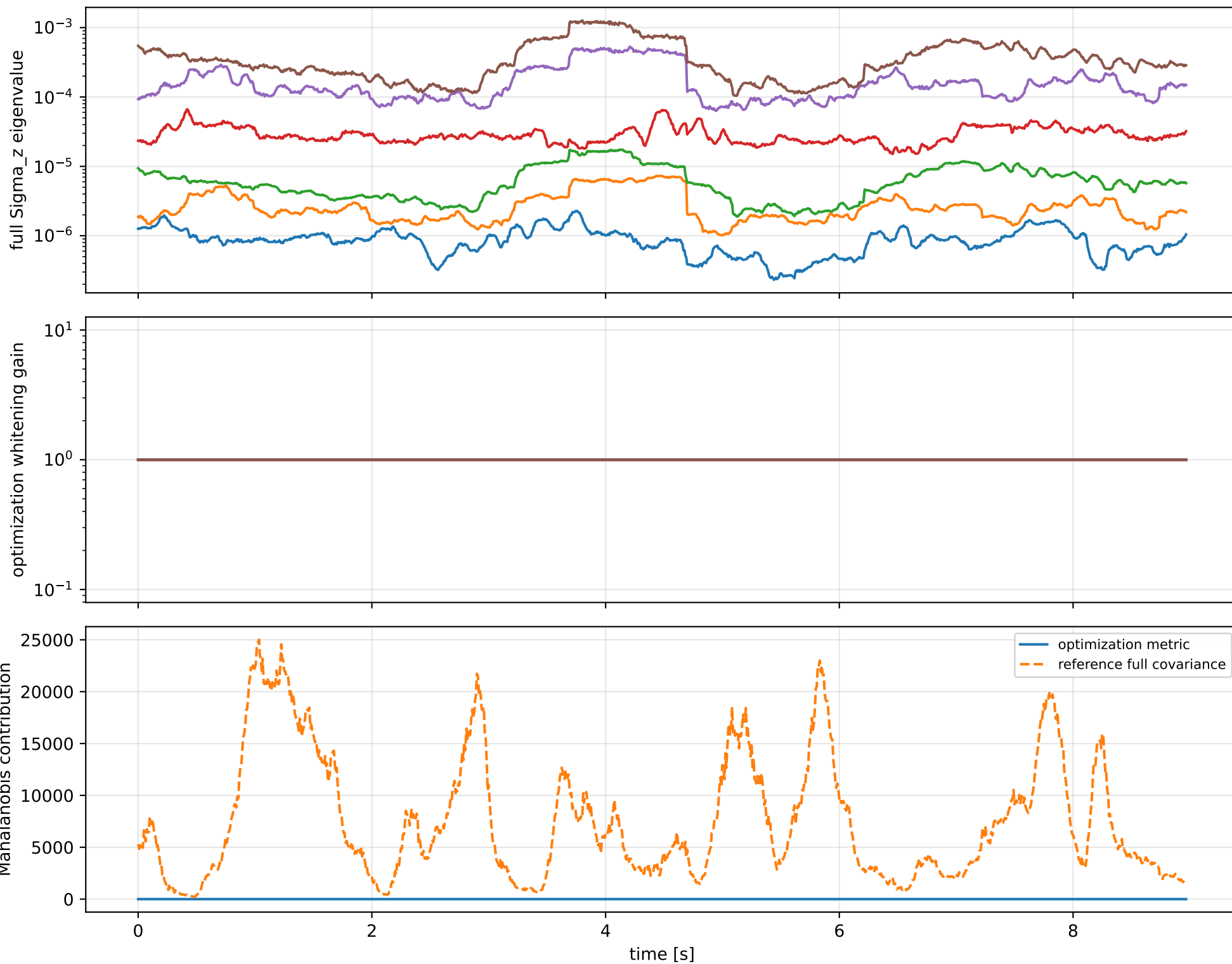
rotor lag cell: (0.175582647, 0.175611973] s

representative: 0.17559731 s

solver_standard_gauge_least_squares: ridge spectrum (rank 12, nullity 2, ||Jv||=4.69e-13)



solver_standard_gauge_least_squares: optimization vs reference covariance



solver_standard_gauge_least_squares: wrench / closure (force max $7.67\text{e}+51$, torque max $4.09\text{e}+49$)

